The Pandrosion Quotient in Quantum Mechanics: Resolvent, Level Repulsion, and a Vanishing Identity

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Abstract

For a quantum Hamiltonian H with characteristic polynomial $P(z) = \det(zI - H)$ and energy eigenvalues $E_1, \ldots, E_d$, the Green's function (resolvent trace) is the Pandrosion Cauchy transform: $G(z) = \mathrm{Tr}(zI - H)^{-1} = P'(z)/P(z)$. The density of states is $\rho(E) = -(1/\pi)\,\mathrm{Im}(P'/P)$, the spectral rigidity is the tidal field $T = E_P/P^2$ (Paper 26), and level repulsion is governed by $P'(E_k) = \prod_{j\neq k}(E_k - E_j)$ (Paper 35). We prove a *quantum vanishing identity*: $\sum_{k=1}^{d} 1/P'(E_k) = 0$, stating that the sum of inverse level repulsions at each eigenvalue vanishes for any quantum system. All results are verified on 200 random Hamiltonians.

1 The Green's function as Pandrosion Cauchy transform

Theorem 1.1 (Green–Pandrosion identity). *For any $d\times d$ Hermitian matrix H with eigenvalues $E_1, \ldots, E_d$ and characteristic polynomial $P(z) = \det(zI - H)$:*

$$\boxed{G(z) := \mathrm{Tr}(zI - H)^{-1} = \frac{P'(z)}{P(z)} = \sum_{k=1}^{d} \frac{1}{z - E_k} = \sum_{k=1}^{d} \frac{Q(E_k, z)}{P(z)}.} \tag{1}$$

Proof. $G(z) = \mathrm{Tr}\ \mathrm{diag}((z - E_k)^{-1}) = \sum 1/(z - E_k) = (\log P)' = P'/P$. The per-quotient form uses $Q(E_k, z)/P(z) = 1/(z - E_k)$. □

The Green's function is the central object of quantum scattering theory, condensed matter physics, and many-body theory. It is the Pandrosion Cauchy transform (Paper 34).

2 Density of states

Proposition 2.1 (Spectral density via Pandrosion). *The density of states is:*

$$\rho(E) = -\frac{1}{\pi}\,\mathrm{Im}\ G(E + i0^+) = -\frac{1}{\pi}\,\mathrm{Im}\,\frac{P'(E + i0^+)}{P(E + i0^+)} = \frac{1}{d}\sum_{k=1}^{d} \delta(E - E_k). \tag{2}$$

With Lorentzian broadening η:

$$\rho_\eta(E) = -\frac{1}{\pi}\,\mathrm{Im}\,\frac{P'(E + i\eta)}{P(E + i\eta)} = \frac{1}{\pi}\sum_{k=1}^{d} \frac{\eta}{(E - E_k)^2 + \eta^2}. \tag{3}$$

Numerical verification confirms that the peaks of ρ_η coincide with the eigenvalues E_k to within η.

3 The quantum vanishing identity

Theorem 3.1 (Quantum vanishing identity)**.** *For any Hermitian matrix H with simple eigenvalues $E_1, \ldots, E_d$:*

$$\boxed{\sum_{k=1}^{d} \frac{1}{P'(E_k)} = \sum_{k=1}^{d} \frac{1}{\prod_{j\neq k}(E_k - E_j)} = 0.} \tag{4}$$

Proof. This is the Pandrosion vanishing identity (Paper 30), applied to the energy spectrum. □

Remark 3.2 (Physical interpretation). $P'(E_k) = \prod_{j\neq k}(E_k - E_j)$ measures the "level repulsion" at eigenvalue E_k: it is the product of all energy gaps from E_k. The identity states that *the sum of inverse level repulsions vanishes*—a fundamental spectral constraint that any quantum spectrum must satisfy.

Verification: 200/200 random Hamiltonians ($d = 3, \ldots, 9$) satisfy $|\sum 1/P'(E_k)| < 10^{-10}$.

4 Level repulsion and spectral determinant

From Paper 35 (random matrices):

$$\text{(GUE density)} \propto \prod_{k=1}^{d} \sqrt{E_P(E_k)} \cdot e^{-d\sum E_k^2/2}, \tag{5}$$

where $E_P(E_k) = P'(E_k)^2$. For a fixed Hamiltonian H, the quantity $|P'(E_k)|$ measures the *isolation* of level E_k:

- Large $|P'(E_k)|$: level is well-separated (strong repulsion).
- Small $|P'(E_k)|$: level is near-degenerate (weak repulsion).

The spectral determinant is the Pandrosion discriminant:

$$D^2 = (-1)^{d(d-1)/2} \prod_{k=1}^{d} P'(E_k). \tag{6}$$

5 Spectral rigidity and the tidal field

Proposition 5.1. *The spectral rigidity—which controls the fluctuations of linear statistics $\sum f(E_k)$—is the Pandrosion tidal field (Paper 26):*

$$T(z) = \sum_{k=1}^{d} \frac{1}{(z - E_k)^2} = \frac{E_P(z)}{P(z)^2} = -G'(z). \tag{7}$$

Regions of large T (near eigenvalues) are "spectrally soft"; regions of small T are "spectrally rigid": the eigenvalues are locked in place by mutual repulsion.

6 Perturbation theory via Pandrosion quotients

For $H = H_0 + \varepsilon V$, second-order perturbation theory gives:

$$\delta E_k^{(2)} = \varepsilon^2 \sum_{j\neq k} \frac{|V_{jk}|^2}{E_k - E_j}. \tag{8}$$

The energy denominators $1/(E_k - E_j)$ appear in the partial-fraction expansion of the function $(z - E_k)/P(z) = 1/Q(E_k, z)$. Thus perturbation corrections are *partial fractions of the inverse Pandrosion quotient.*

7 Partition function

The partition function $Z(\beta) = \mathrm{Tr}(e^{-\beta H}) = \sum e^{-\beta E_k}$ is connected to the Green's function via the Laplace contour:

$$Z(\beta) = \frac{1}{2\pi i} \oint \frac{P'(z)}{P(z)}\, e^{-\beta z}\, dz = \sum_{k=1}^{d} \mathrm{Res}_{z=E_k} \frac{P'(z)}{P(z)}\, e^{-\beta z} = \sum_{k=1}^{d} e^{-\beta E_k}. \tag{9}$$

8 Conclusion: complete dictionary

Quantum physics	**Pandrosion**
Hamiltonian H	Matrix with char. poly P
Energy levels E_k	Roots of P
Green's function $G(z)$	P'/P (Paper 34)
Density of states $\rho(E)$	$-(1/\pi)\,\mathrm{Im}(P'/P)$
Level repulsion	$P'(E_k) = \prod(E_k - E_j)$ (Paper 35)
Vanishing identity	$\sum 1/P'(E_k) = 0$ (Paper 30)
Spectral determinant	$D^2 = \prod P'(E_k)$ (Paper 22)
Partition function	$\oint (P'/P) e^{-\beta z}\, dz$
Perturbation theory	Partial fractions of $1/Q(E_k, z)$
Spectral rigidity	Tidal field $T = E_P/P^2$ (Paper 26)

The quantum vanishing identity $\sum 1/\prod(E_k - E_j) = 0$ is a universal constraint on any quantum spectrum with simple eigenvalues.

References

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